Introduction

The aim of the project is to acquire**overview of how data science works**, basic concepts and techniques of data analysis, understand how they work and gain intuition for their appropriate application for the purpose of discovering knowledge in data. They will also gain an idea of what questions we can answer and apply using data analysis**basic machine learning approaches**Emphasis is placed on data analysis and preprocessing, the use of machine learning methods, and methods of their evaluation and comparison.

The project is being developed**in pairs**in acceptable quality. The solution uses a programming language**Python**and available libraries for data science like**pandas, numpy, scipy, statsmodels, scikit-learn,**etc... At each stage and activity, an executable is delivered**Jupyter Notebook**to AIS, which contains all the transformations performed on the data with appropriate documentation. The submitted notebook must contain not only the code, but also its results (calculated values, statements, visualizations, etc.) along with a comment on the obtained results and the resulting decisions for the next steps of the data process. The ability to communicate well and present relevant results is an important component of the assessment.

For each stage in the notebook you submit, please indicate**percentage of work**members of the couple.

Data

https://drive.google.com/drive/folders/1DJbqyN_TtSfHdq4_qQl6dbqsVmLGlWO1?usp=sharing ***(each pair has one dataset under the number you have for the exercise)***

Additional information about the data:explanation of selected sensor values

Oxygen saturation (eng.**oxygen saturation**) is a key indicator of the proper functioning of the respiratory and circulatory systems. If its value drops to a critically low level, it can signal life-threatening conditions such as hypoxemia, respiratory failure or serious infections. In such cases, rapid intervention is essential. Traditional monitoring is performed using pulse oximeters, which, however, can be affected by noise, motion artifacts or have limitations in some clinical situations.

Modern approaches based on machine learning bring the possibility of more accurately estimating and predicting critical values of oxygen saturation (**critical oxygen saturation estimation**). Models can use multimodal data such as heart rate, respiratory rate, blood pressure, and sensor signals. By training on diverse data, it is possible to identify early warning signs of desaturation, filter out noise, and provide early warnings before saturation drops below a safe limit.

The aim of this assignment is to familiarize yourself with the issue of oxygen saturation monitoring, understand the benefits of artificial intelligence, and propose a solution that could contribute to improving critical care and reducing the risks associated with undetected hypoxemia.

Project assignment

Critical oxygen saturation estimation

The QUEST

Each pair will work with an assigned data set starting from week 2.**Your task**is to predict the dependent variable values"**oximeters**"(predicted variable) using machine learning methods. You will have to deal with several problems found in the data such as data formats, missing values, outliers, and many more.

Expected**output**of the project is:

1. **best model**machine learning;
2. **data pipeline**for its construction based on input data.

Phase 1 - Exploratory Analysis: 15% = 15 points

1.1 Basic description of the data along with its characteristics (5b)

EDA with visualization

- (A-1~~b) Analysis of data structures such as files (structures and relationships, number, types, ...), records (structures, number of records, number of attributes, types, ...)~~
- (B-1b) ~~Analysis of individual attributes: for selected significant attributes (min. 10), analyze their distributions and basic descriptive statistics and whether they meet the prescribed conditions and range of measured values.~~
- ~~(C-1b) Pairwise data analysis: Identify relationships and dependencies between pairs of attributes.~~
- ~~(D-1b) Pairwise data analysis: Identify dependencies between **predicted** variable and other variables (potential predictors).~~
- ~~(E-1b) Document your initial thoughts on solving the project assignment, e.g. are some attributes dependent on each other? On which attributes does the predicted variable depend? Is it necessary to combine records from multiple files?~~

1.2 Problem identification, data integration and cleaning (5b)

- ~~(A-2b) Identify and initially resolve problems in the data, e.g.: inappropriate data structure, duplicate records that may arise after certain data transformations, inconsistent formats, missing values, outliers. There may be other problems in the data not listed here, or fewer problems than listed.~~
- ~~(B-2b) Checking the correctness of the data~~
  - ~~whether they contain abnormal values~~
  - ~~whether they contain illogical data relationships that are a result of data collection and data annotation~~
- (C-1b) ~~Outlier detection, try to solve the problem with at least 2 techniques~~
  - ~~removing~~ biased ~~or outlier observations~~
  - ~~replacing the outlier with the distribution boundary values (e.g. 5%, 95%)~~

1.3 Formulation and statistical testing of hypotheses about data (5b)

- ~~(A-4b) Formulate **two hypotheses** about the data in the context of the given prediction task. Verify the formulated hypotheses with appropriately selected statistical tests.~~
  ~~Example of wording:~~

  FiO_2 has a higher value on average in the state with oximeters as without it.
- ~~(B-1b) Verify that your statistical tests have sufficient support from the data, i.e. whether they have sufficiently strong statistical power.~~

In the submitted report (Jupyter notebook), you should answer the following questions:
Is the data in a suitable format for further processing? What problems are occurring in it? Are some attributes taking on inconsistent values? How do you solve these problems you have identified?

The report is submitted in the 5th week of the semester.. The pair presents the completed phase to their trainer in a Jupyter Notebook as needed for the exercise. In the notebook, indicate **percentage of work** members of the couple. The report will then be submitted electronically. **one member of a pair** into the system **AIS** until Sunday **26.10.2025 23:59**.

Phase 2 - Data Preprocessing: 15 points

The report is submitted in the 7th week of the semester.. The pair presents the completed phase to their trainer in a notebook as needed for the exercise. Indicate the percentage of work of the pair members. The report will then be submitted electronically.**one member of a pair**into the system**AIS**until Sunday**09.11.2025 23:59**.

Phase 3 - Machine Learning: 20 points

The report is submitted in the 10th week of the semester. The pair presents the completed phase to their trainer in a Jupyter Notebook for the exercise. In the notebook, indicate the percentage of work of the pair members.The report will then be submitted electronically.**one member of a pair**into the system**AIS**until Sunday **30.11.2025 23:59**.

Exercise activities: 10 points

The report is submitted in the 12th week of the semester. The pair presents the completed phase to their trainer in a Jupyter Notebook for the exercise. In the notebook, indicate the percentage of work of the pair members.The report will then be submitted electronically.**one member of a pair**into the system**AIS**until Sunday **14.12.2025 23:59**.